Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Normal force

Meaning in my words How would you explain normal force to a friend?	**How can you tell?** A crate rests on a ramp. Which way does the ramp push the crate?
Example or sketch How would you draw the normal force on a book on a level table?	**Non-example and why** Someone draws the normal force along the ramp. What would you rotate?

Weight

Meaning in my words How would you explain weight to a friend?	**How can you tell?** When a book rests on a table, which object supplies its weight?
Example or sketch Using 10 N/kg, what arrow represents the weight of a 3 kg book?	**Non-example and why** A book slides down a ramp. Should its whole weight arrow point along the ramp?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Component

Meaning in my words How would you explain component to a friend?	**How can you tell?** Which part of the block's weight pulls it down the ramp?
Example or sketch How could you split one downward weight arrow using ramp axes?	**Non-example and why** Should the downhill component be added to the full weight as another force?

Acceleration direction

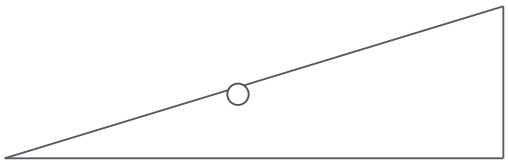
Meaning in my words How would you explain acceleration direction to a friend?	**How can you tell?** A cart rolls uphill on a frictionless ramp. Which way does it accelerate?
Example or sketch At the top of its trip, draw the acceleration arrow just as the cart stops.	**Non-example and why** "The cart moves uphill, so acceleration points uphill." Which arrows got mixed up?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw the real weight once, then draw its two components in a separate box. Circle the component along the ramp.

Real forces



Components

My marks mean

On assignment: Component

SECTION E / BACK / FINAL MEETING ONLY Name _____

A navigation graphic lists a gravitational force AND its two components as three separate pulls on XJ.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Navigation engineer**

Build an overlay using the component view. Indicate the full arrow it replaces before releasing the values to the calculator.

☐ **Map graphics editor**

Sketch two camera views: the real pull, then its components. Give viewers a caption that prevents counting the same pull twice.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.